

Case Study – Quantitative Analyst – Parametric Insurance

About InRisk Labs

InRisk Labs develops innovative parametric insurance products that protect businesses against weather and climate risks. Our products use objective weather data to provide transparent, fast, and scalable insurance solutions.

As a Quantitative Analyst, you will be expected to work with climate datasets, design weather indices, price products, and develop commercially viable insurance solutions. This case study is intended to simulate a real product development problem at InRisk Labs.

Business Problem

A financial institution finances 3 kVA rooftop solar systems for residential customers across India. The lender's concern is that prolonged periods of adverse weather can significantly reduce electricity generation, resulting in lower savings for homeowners and potentially affecting loan repayments.

They have approached InRisk Labs to develop a parametric insurance product that compensates customers whenever weather conditions lead to unusually low solar energy generation.

Product Specifications

A customer has installed a **3 kVA grid-connected rooftop solar system** at the Pincode: 380006

Technical Assumptions

Assume the rooftop solar system has the following characteristics:

Parameter	Value
Installed Capacity	3 kVA
Performance Ratio (PR)	80%
AEP50 (Annual Energy Production at P50)	4,600 kWh/year
Electricity Tariff	₹5.75 per kWh
Cost of Installation	₹2,50,000

Business Context

A residential customer has financed a rooftop solar system through a bank.

The customer's expected electricity savings are based on the expected annual energy generation (AEP50).

However, unusually poor weather conditions may significantly reduce electricity generation, lowering annual savings.

The financial institution wishes to insure customers against this reduction in financial benefit using a parametric weather index.

Your Objective

Design a **parametric insurance product** that compensates the customer when weather conditions lead to significantly lower-than-expected electricity generation.

Your product should:

- Minimize basis risk.
- Be transparent and easy to explain.
- Be commercially viable.
- Be actuarially priced.

Data Acquisition

You are required to download the relevant historical climate data for the policy location from ERA5-Land.

- **Location:** Pincode 380006
- **Historical Period:** 2005–2024

You are free to select the climate variable(s) that you consider most appropriate for designing and pricing the proposed parametric insurance product.

Tasks

Task 1 – Weather Index Selection

Identify the weather variable(s) that best explain reductions in rooftop solar generation.

Explain:

- Why you selected the variable(s).
- Why alternative variables were not selected.
- Any assumptions made.

Task 2 – Product Design

Design a parametric insurance product.

Clearly define:

- Weather index
- Trigger
- Payout structure
- Maximum payout
- Any assumptions or simplifications.
- Limitations of your pricing methodology.

The payout should compensate the customer for reduced financial savings due to adverse weather conditions. You may make reasonable assumptions where information is unavailable.

Task 3 – Pricing

Using historical ERA5-Land data:

- Back-test your proposed product and estimate the annual payout for each historical year.
- Estimate the expected annual loss cost (burn cost) and recommend an appropriate commercial premium.

Task 4 – Product Critique

Suppose this product has been sold to 10,000 customers.

Identify three situations where the product may fail to provide a satisfactory customer experience.

For each situation:

- explain why it occurs,
- estimate its potential impact,
- propose one improvement.

Submission Guidelines

- Submit your work in a 3-4 page report including the approach you took and why you opted for that specific approach
- Share all your workings and codes
- You need to specify wherever AI has been used and share your prompts along with the submission.

REMEMBER, THERE IS NO CORRECT OR INCORRECT SOLUTION FOR THIS CASE STUDY – WE ARE MORE CONCERNED WITH YOUR THOUGHT PROCESS AND ABILITY TO USE ALL RESOURCES AT DISPOSAL THAN YOUR ACTUAL SOLUTION.